Introduction:
The TW4 energy recovery ventilator is installed using a 6 inch (150 mm) nominal diameter SDR-35 pipe through a wall. The outer diameter of the pipe is actually about 159 mm.

This document describes a sound attenuator that blocks a high degree of both fan noise and also exterior noise such as traffic. It attaches to the exterior side of that through wall pipe. The 3d printed “noise splitter” (not shown), which is used on the WM12, only splits the fan noise from the air stream. The “noise splitter” allows exterior noises from e.g. traffic through. This heavy duty attenuator, in contrast, attenuates both fan noise and exterior noise such as traffic, by about 22 dBa.

It is made entirely from hardware store parts/materials. It is extremely durable, including in the face of UV exposure. Although not cheap, it’s made of the same building materials buildings are normally made of, so that’s just the price of good quality standard building materials, basically.

The fan mount provisions are different. It is a circular 3d printed socket for the fans, surrounded by neoprene foam weather stripping. It is not really part of the attenuator but it is not shared in common with the other approach so it is described/included here..

The parts you will need to buy/have ready are as follows:

Part	quantity/length	Approximate cost each (CAD, divide by 1.3 for USD)
End cap	1	15
Main pipe 150 mm SDR-35	50 cm +/- 8 cm	13
T fitting	1	50
100 mm pipe SDR-35	20 cm	5
Fiberglass (dimensions not critical, extras can just be stuffed in the pipe etc.) the best stuff is the wrapping around insulated supply side interior flex duct. Almost any other fibrous material will work, but fiberglass is mold and fire proof and has no significant smell.	240 +/- 1.5 cm by 45 +/-2 cm	8
Reducer bushing	1	20
Sum		111

Table 1.

Instructions:

1. The hardest part is to saw the shoulder off the reducer bushing, the result of which is shown in one of the pictures. I used an oscillating saw with circular blade, a drywall type blade might work better. A hand saw can work, it's just a bit difficult.
2. Cut the main pipe. A good method to get a square end is to take two sheets of paper, tape them together end to end carefully so they are straight/the edges are parallel, then wrap it around the pipe and tape it in place (ideally with masking tape so it's easy to remove). Then mark the pipe along the edge of the paper with a sharpie in the right position along the pipe, remove the paper and cut it with a hand saw. You can also use packaging tape or masking tape with the adhesive on the outside, that may be easier.
3. It is helpful to chamfer or sand slightly the edge of the pipes, so it is not sharp. This makes it easier to put in the fitting. This applies to the 100 mm and 150 mm pipes alike.
4. The main pipe can be spray painted white if desired, at this stage, I haven't tried painting it but if you clean it and use a paint that is suitable for plastic and exterior use it should work fine. You can always do it another day, too.
5. Trim the gaskets. This requires a sharp knife, a small (10 mm wide blade) boxcutter with a new blade is good. Trim it to within 1 mm of the yellow part. If you don't trim it enough, it is too hard to push the pipe in, too much and the pipe will not be a very tight fit and

could come out. If you go too far that would get complicated, so be conservative, try to fit the pipe in and trim more if you need to. Grease also helps a lot to get the pipe in.

6. The gasket on the T fitting where it will connect to the ERV is the most important, we don't want to push too hard on the pipe in the wall, but we also don't want it to come off or rotate under the influence of wind or gravity. The upper part is heavier than the lower part so this could happen if it's not on there securely and it got a good push from a strong wind. I have no solution for this right now except making the sound trap shorter. I recommend not putting grease on this area, too. Water may be used to make it easier to put on, it will evaporate after. Kerosene or diesel would also probably work better, but smells.
7. Cut the 100 mm pipe, 20 cm long. Use the paper trick to get a square end.
8. Dismantle the supply side interior flex duct to get at the fiberglass blanket. Cut the fiberglass with scissors to approximate size, if in doubt just leave extra, you can stuff it in relatively easily, the main thing is that it must not block the 100 mm pipe end. If you put too much in, you can tear some out later. The dimensions given are just about right to make a sheet which can be simply rolled up tightly and stuffed in.
9. Put the end cap on the main pipe
10. Rollup and stuff the fiberglass in. It is actually ideal if the end facing the 100 mm pipe is rough/uneven, so the density changes gradually.
11. Insert the 100 mm pipe in the reducer bushing
12. Insert the reducer bushing into the T fitting
13. At this point, put the fans in the bottom of the reducer bushing and run the wires, they cannot be as easily run after putting the main pipe on the T fitting, so it should be done now.
14. Attach the wire extension to the fan plug.
15. Push the main sound trap pipe into the T fitting.
16. Inspect it, make sure the fiberglass is not blocking the end of the 100 mm pipe. There should be at least 4 cm of clearance between the end of the fiberglass and the end of the 100 mm pipe.

Now the assembly is ready to attach to the pipe on the exterior wall. It is a bit awkward to attach the wires while holding it at the same time, unfortunately.

When installing it, I therefore recommend that the wires just be placed in the pipe, and the attenuator attached to the outside securely. The heat exchanger would be added later.

Taping the attenuator to the exterior pipe may be a good idea, depending on how tight the gasket tha connects the main through wall pipe to the T fitting fits. We don't want it to come off or rotate under the force of strong winds and/or gravity over time. With ungreased rubber in contact with clean plastic there should be no gradual motion with time. It also will help if the main pipe is very close to vertical.

Pictures:
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POWER ARMOR 16 PRO

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Peso neto: 9.4 kg (20.9 lb)
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Figure 8. Fans not shown, only the printed fan mount piece